DATA MINING PROJECT

Semester Research Project

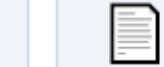
Apply real data mining techniques to a real-world problem — and publish a research paper.



Team Size

1-3 members per team

Recommend to Work as a Team since a lot of tasks to do until the completion and semester time is limited.



Final Output

Conference-standard paper (with literature review, methodology, experiment, result, conclusion and citation)



Data Source

Kaggle · UCI ML Repository · Thailand Open Government Data · Your own collected data

Note that: If the data is public, there might be some research publications using it and you need to review and beat them.



Kick-off

Make 2-3 slides
Bring brief information and screen out only information Ajarn needs to know and approve,
Bring sample of data

Request timeslot for consult during 14-16Oct.

Start brainstorming now — bring your idea to discuss after the midterm break (Day 08)

Choose Your Research Track



ARM

Association Rule Mining

Discover hidden co-occurrence patterns in transaction data

e.g. Market basket analysis · Co-purchased products · Behavior co-occurrence



CLUSTERING

Unsupervised Clustering

Find natural groupings without predefined labels

e.g. Customer segments · Topic modeling · Geo-spatial clusters



CLASSIFICATION

Classification

Predict which category a new observation belongs to

e.g. Disease detection · Spam / fraud detection · Churn prediction



PREDICTION

Regression / Prediction

Forecast continuous or numerical outcomes from features

e.g. Sales forecasting · Rainfall prediction · Score estimation

Class of 2024 — Project Showcase

Categorized by DM Task Type

Link => https://drive.google.com/drive/folders/1YKw3J5yEjL5PEcs3IHywfDIKrSMnmAC-

ARM	CLUSTERING	CLASSIFICATION	PREDICTION
Mining Football Players: Data-driven Scouting Approach	Lifestyle Factors for Mental & Physical Well-Being	Detecting Phishing URLs	Remote Work Impact on Mental Health
Music → Emotions: Personalized Support Playlist	Tourist Spending Behavior Segmentation & Prediction	TV Show Cancellation Prediction	Song Popularity Prediction
GitHub Repo Activeness & Related Stats	Game Genre Popularity & Player Behaviour	Loan Approval Classification	Air Pollution Prediction in Urban Areas
Climate Change Effects on Species Extinction	Human Rights Index Analysis (2018–2021)	Colon Cancer Gene Expression (DEGs + ML)	Rainfall Prediction in Thailand
	Personal Finance Management	NLP-Driven Emotion Prediction (Text Mining)	Kickstarter Project Success Prediction

These projects were submitted in 2024. Your project should be equally or more interesting — unique or reused dataset, clear research question, and publishable insights.

What Makes a Great Project?

✅ DO	❌ AVOID	💡 PRO TIPS
Pick a problem you genuinely care about	Toy or textbook datasets (iris, titanic, etc.)	Combine DM with a domain you know well
Use at least 500 real-world records	Running just one model with no comparison	Thai or ASEAN datasets are underrepresented → higher novelty
Apply proper evaluation (accuracy, lift, silhouette…)	Copy-pasting code without understanding it	Add one unexpected variable to your model
Discuss what your results mean for real life	Vague research question ("I want to predict sales")	Visualize results — charts tell stories better than tables
Follow CRISP-DM — show every phase in the paper	Results without explanation or context	Keep a data diary (what you changed and why)
Cite related work (10 references minimum within 5 years)	Team members who only show up on Day 15	Think "would it good enough to submit to the conference?"

📁 Submit to: drive.google.com/drive/folders/1YKw3J5yEjL5PEcs3IHyvfDIKrSMnmAC- · Questions → bring them to Day 08

Choosing Your Project: 6 Traps to Avoid

Goal: a paper good enough to publish, not just code that runs.

1 🚫 Everyone already used this dataset

Popular Kaggle / UCI data = lots of papers exist already.

✅ **Search Google Scholar: "dataset" + task. Beat them, or ask a new question.**

2 🚫 Fake (synthetic) data

Fake data says nothing about real life.

✅ **Use real data. New algorithm? Test it on several datasets.**

3 🚫 "Nobody did this before!"

Someone probably did. Missing it = you repeat their work.

✅ **Try many keywords: Google Scholar, IEEE, ACM, Springer.**

4 🚫 Too few references

No literature = not research.

✅ **10–15 papers in your draft, mostly from the last 5 years.**

5 🚫 Skipping the Project Info

Repeating last year's topics wastes weeks.

✅ **Read this deck + the Class of 2024 showcase before you pitch.**

6 🚫 No plan for data & models

"I'll find data later" = weak method (30% of the score!).

✅ **Pick source, ≥ 500 rows, ≥ 3 algorithms + a baseline.**

Before you pitch: ① Which 10+ papers study this already? ② What is new in our work? ③ Real data, ≥ 500 rows?

Submit Once: Your Public Project Webpage

One link for the whole semester. No more sending files one by one!

1

BUILD IT

Make a free, public webpage for your team.
Google Sites · GitHub Pages · Notion (publish to web)

2

SUBMIT THE LINK ONCE

Post your page URL on Google Classroom by 29 Oct.
Include team name, members and track.

3

KEEP IT UPDATED

Add your new work before each milestone date.
We open your page on that day: that is your submission.

Your page must...

- ✓ Open for anyone with the link: no login
- ✓ Show team names, track and "last updated" date
- ✓ Date-stamp every progress entry
- ✓ Link to big files (paper PDF, code, data)
- ✓ Contain no private or personal data

Page not updated by the milestone date = late (–10% per day). A broken or private link = not submitted.

Project Timeline: What Your Page Must Show

4 milestones from idea to final paper. Update the SAME page each time.

MILESTONE 1
14–16 Oct

Week 8 · by appointment

Pitch your idea

✓ Meet the teacher: 15 min per team
✓ Printed proposal (1–2 A4 pages)
✓ Slides (if you have any)
→ Copy your proposal onto your page later

MILESTONE 2
27 / 29 Oct

Week 10 · Day 10

Lock your data

✓ Page is LIVE + link submitted
✓ Home + Problem + research question (1 sentence)
✓ Data: source, real, ≥ 500 rows

MILESTONE 3
10 / 12 Nov

Week 12 · Day 12

Show progress

✓ Data cleaned: what you changed
✓ Initial results: baseline + ≥ 2 models
✓ First draft: what is working, what is not, how to solve it

MILESTONE 4
24 / 26 Nov

Week 14 · Day 14

Final submission

✓ Present: 10 min talk + Q&A
✓ Paper PDF: IEEE/ACM, 4–6 pages
✓ Slides, code and dataset linked on the page

⚠ **Late = −10% per day. Final exam: 3 Dec. Your page = your submission: check the link opens in a private browser window.**

Page Blueprint: 7 Sections

Keep it simple. One scrolling page is fine. Charts and screenshots beat walls of text.

1 Home

Title · team members · one-sentence summary · last updated

Add by: M2

2 Problem

Why it matters · research question · related work

Add by: M2

3 Data

Source · size (≥ 500 rows) · columns · cleaning steps

Add by: M2 → M3

4 Method

Algorithms (≥ 3) · baseline · evaluation plan

Add by: M3

5 Progress log

Dated entries: done · stuck · next

Add by: every milestone

6 Results

Tables with mean ± std · charts · what they mean

Add by: M3 → M4

7 Files & references

Paper PDF · slides · code · dataset · 10+ references

Add by: M4

Quick tips

- Free tool, no code needed
- Look at a past team page
- Test the link on your phone

Before every milestone: ① page opens without login ② new section added ③ "last updated" date is correct